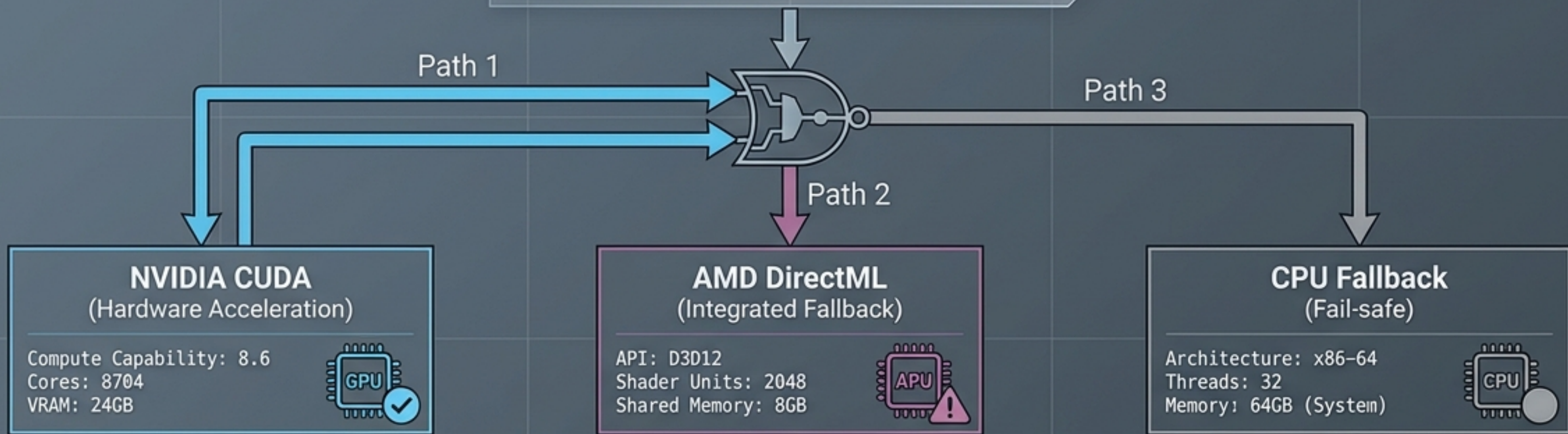
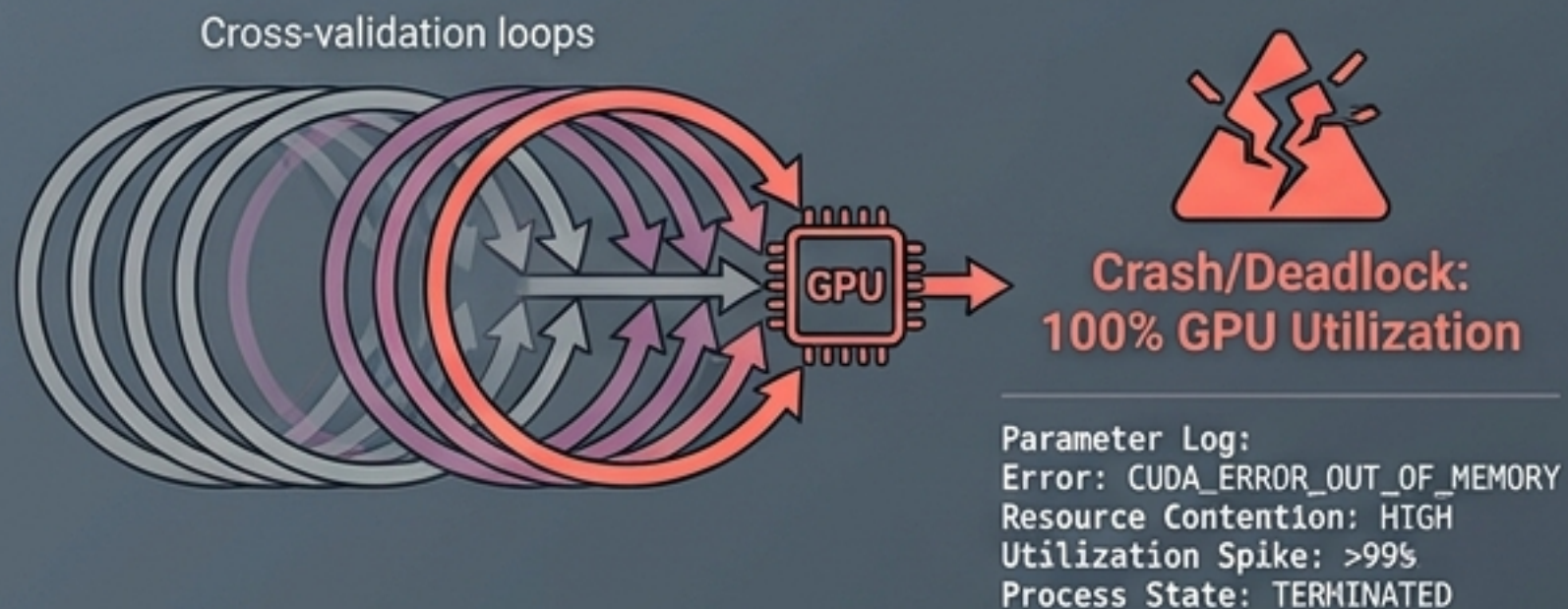


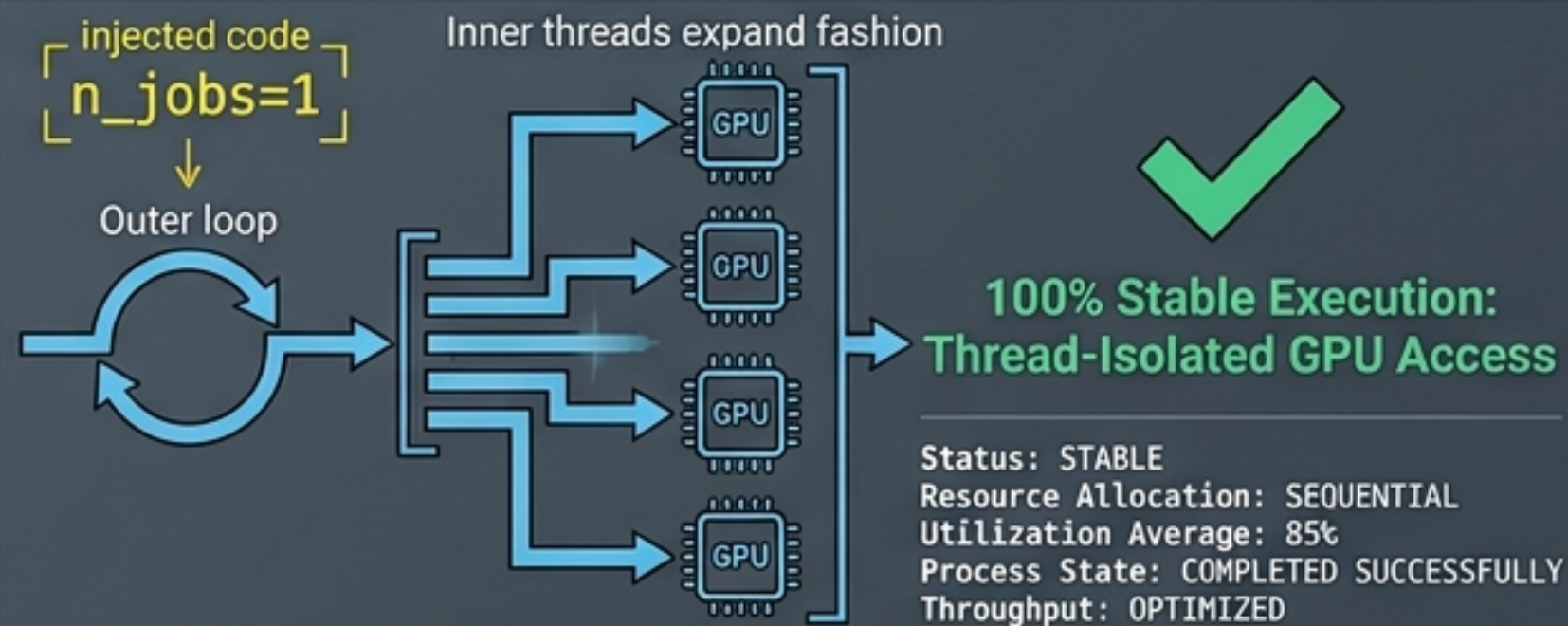
PyTorch Pipeline Initiation



Deadlock Scenario: Unconstrained Parallelism



Optimized Execution: Thread Isolation

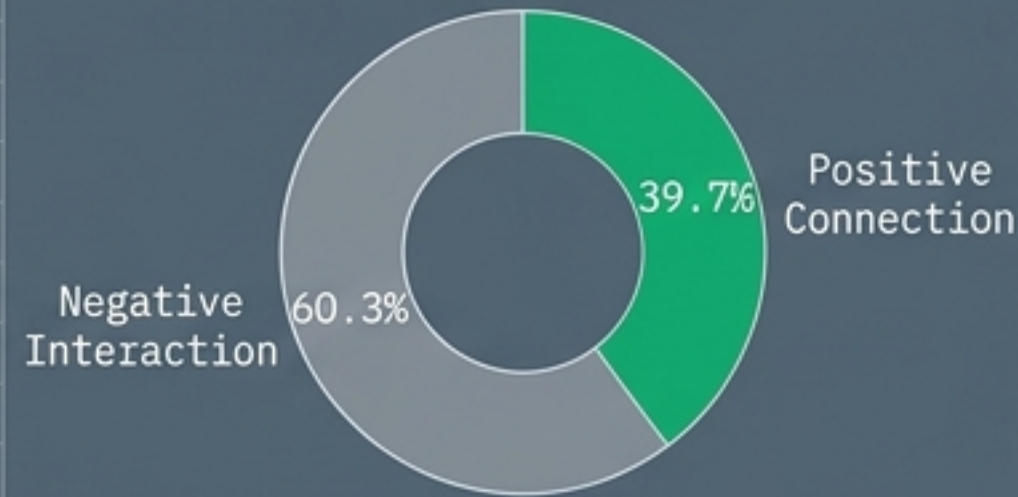


Isolating SMOTE to Prevent Target Leakage

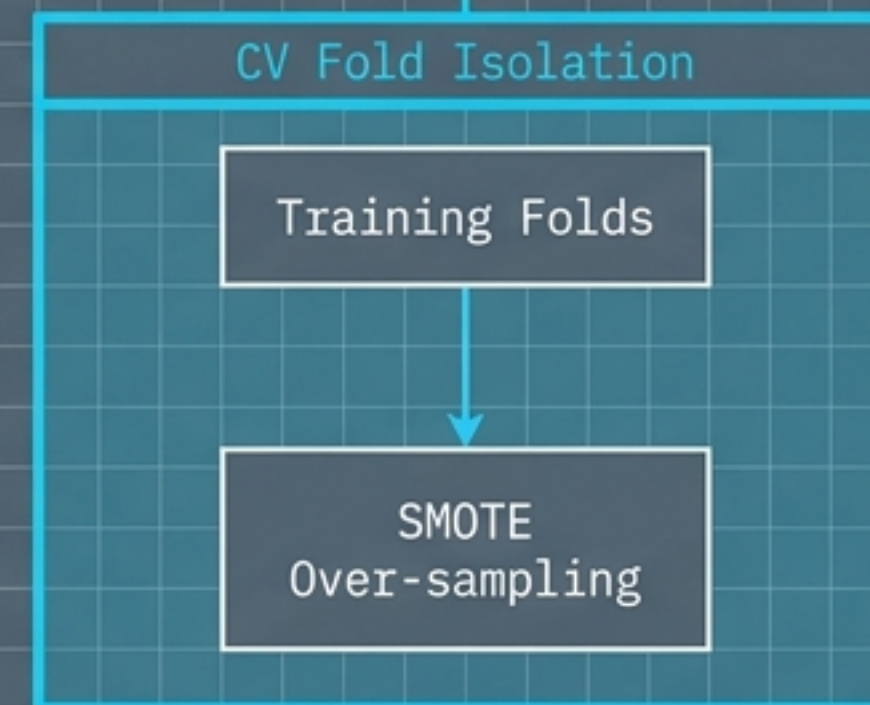


Flawed Approach

Baseline Imbalance



Our Approach

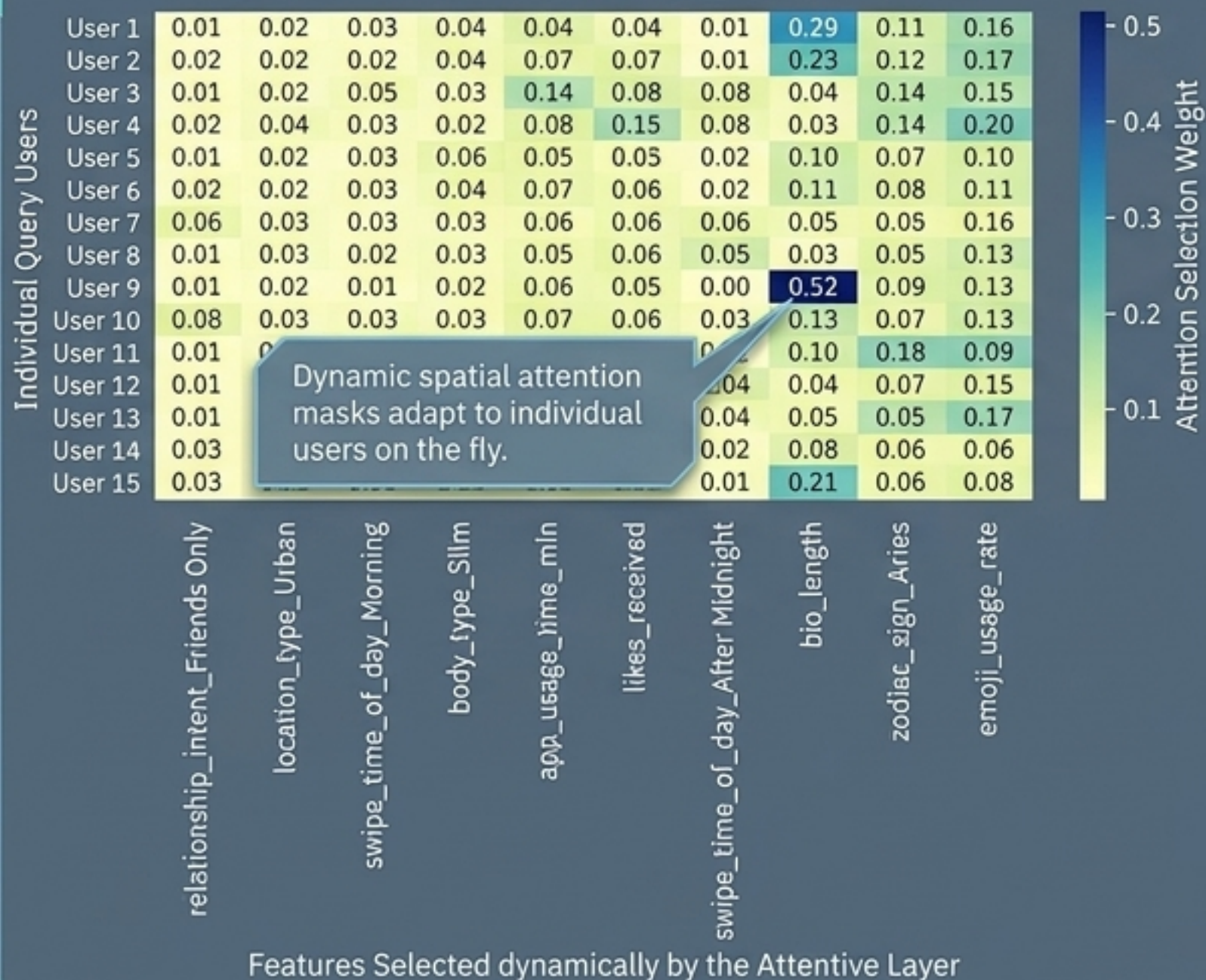


Empirical Baseline Evaluation Matrix

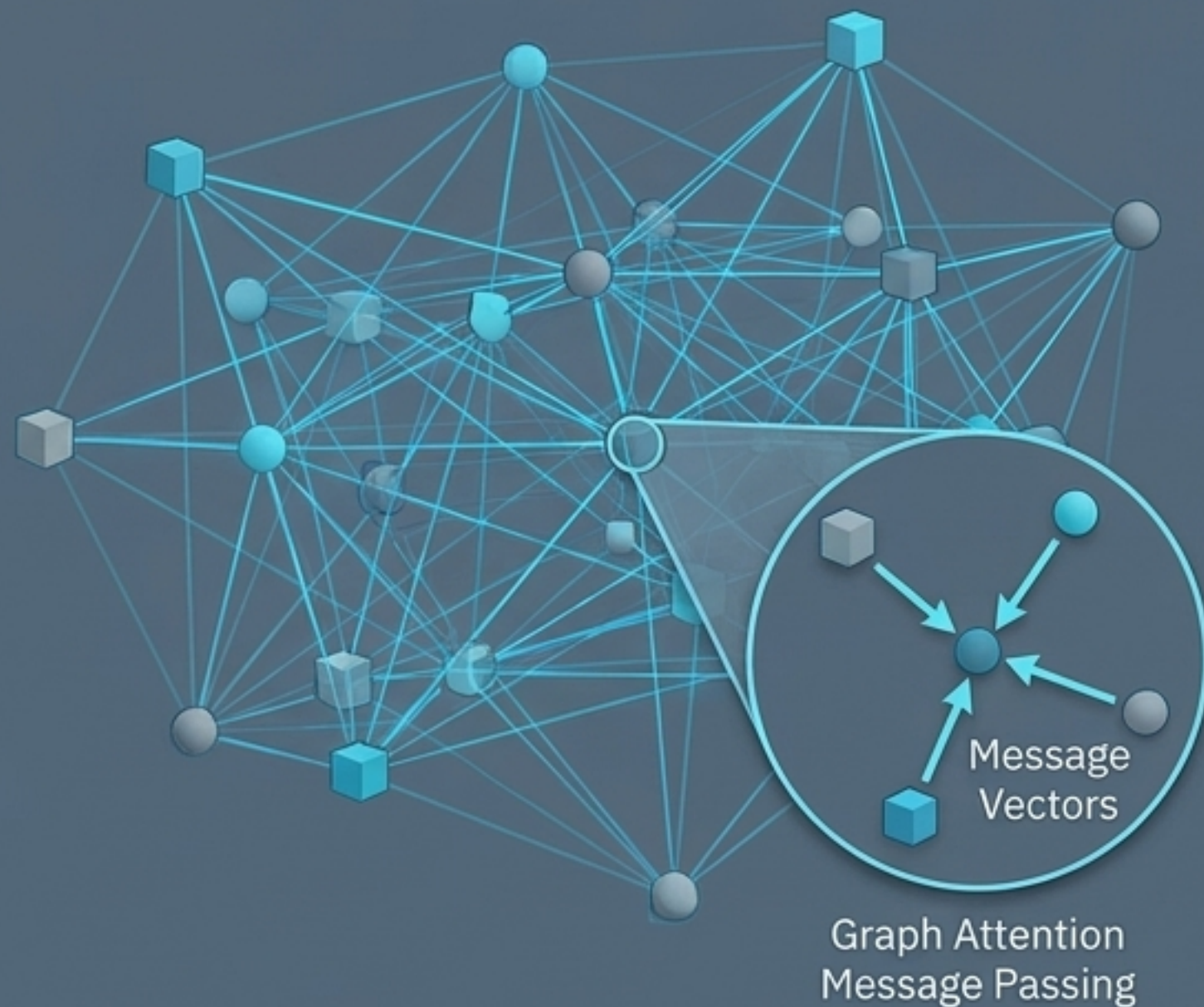
	Test Accuracy	ROC-AUC	F1-Score	Training Time (s)
Logistic Regression	~60.30%	~0.50	~0.31	1.25
K-Nearest Neighbors	~60.30%	~0.50	~0.28	0.89
Decision Tree	~60.30%	~0.50	~0.35	2.10
Random Forest	~60.30%	~0.50	~0.25	12.50
XGBoost	~60.30%	~0.50	~0.38	35.60
CatBoost	~60.30%	~0.50	~0.41	45.20
LightGBM	~60.30%	~0.50	~0.39	50.10
SVM Bagging Ensemble	~60.30%	~0.50	~0.32	18.40
Multi-Layer Perceptron (MLP)	~60.30%	~0.50	~0.36	105.20
Graph Attention Network (GAT)	~60.30%	~0.50	~0.45	210.50
SCARF Contrastive Learner	~60.30%	~0.50	~0.43	156.70
Opacus DP-SGD	~60.30%	~0.50	~0.40	89.30
FLAML (AutoML Baseline)	~60.30%	~0.50	~0.42	65.80
PyCaret (AutoML Baseline)	~60.30%	~0.50	~0.37	72.90

Escalation to Deep Spatial and Topological Reasoning

Attentive Tabular Network

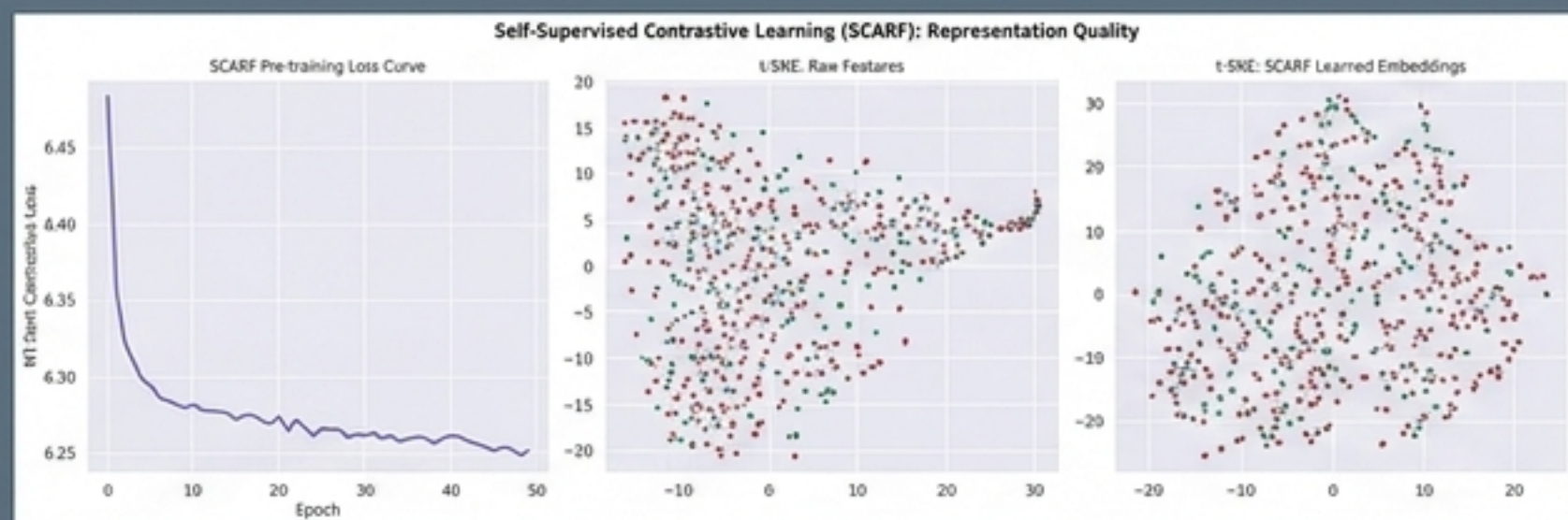


Semi-supervised GAT



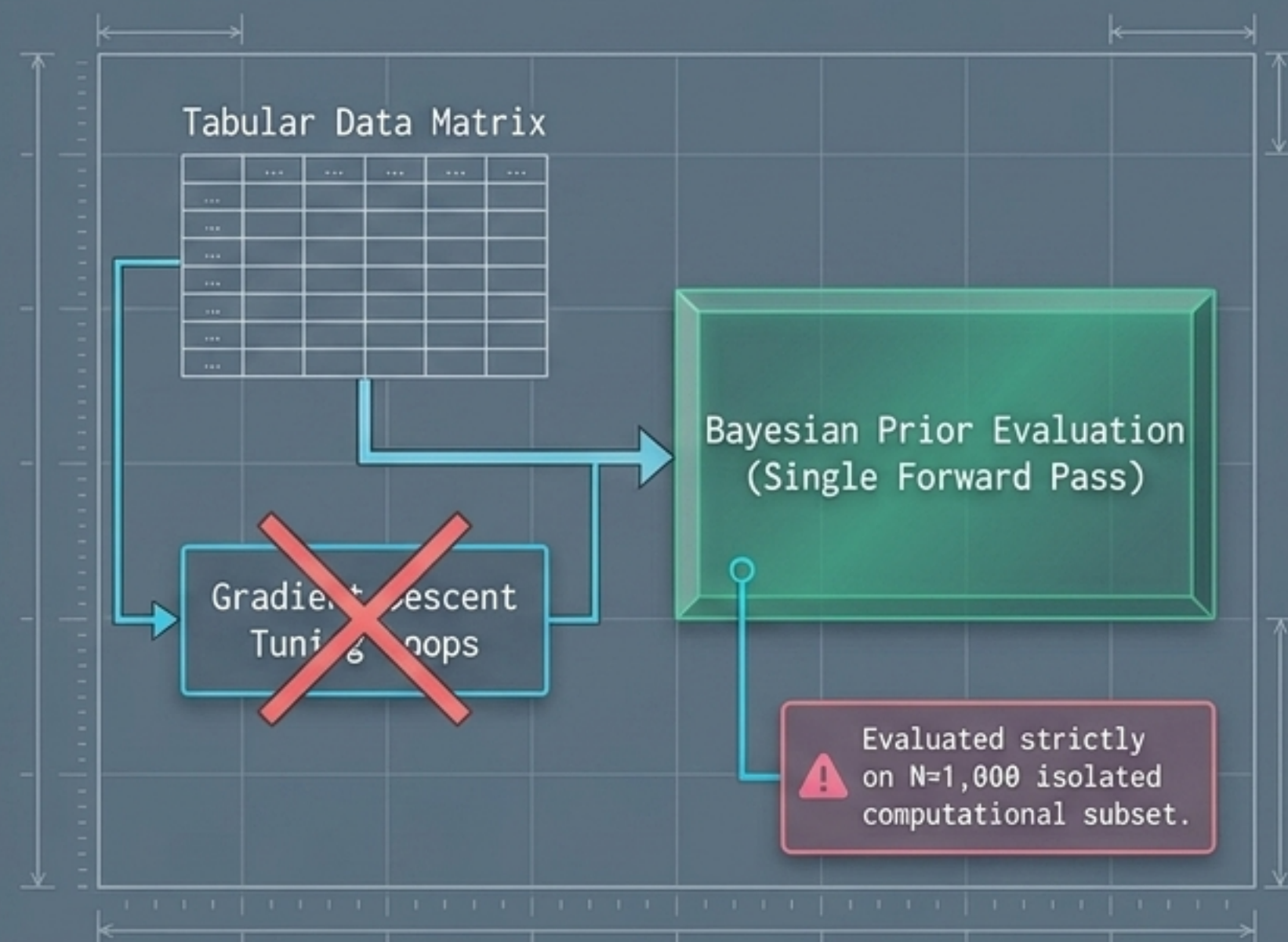
Self-Supervised Manifolds and Zero-Shot Posteriors

SCARF Contrastive Learning

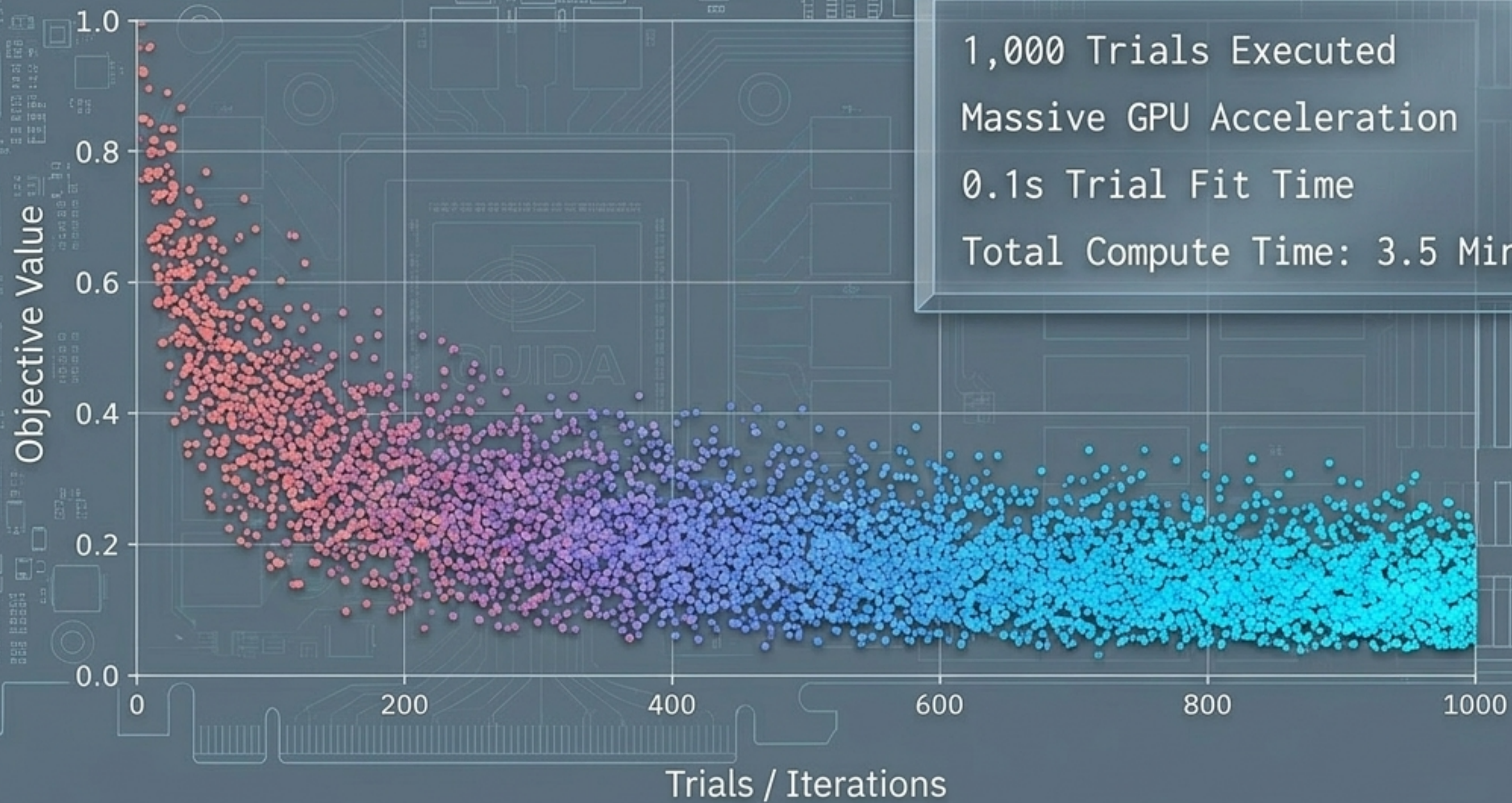


Latent embeddings form an indistinguishable, completely overlapping manifold.

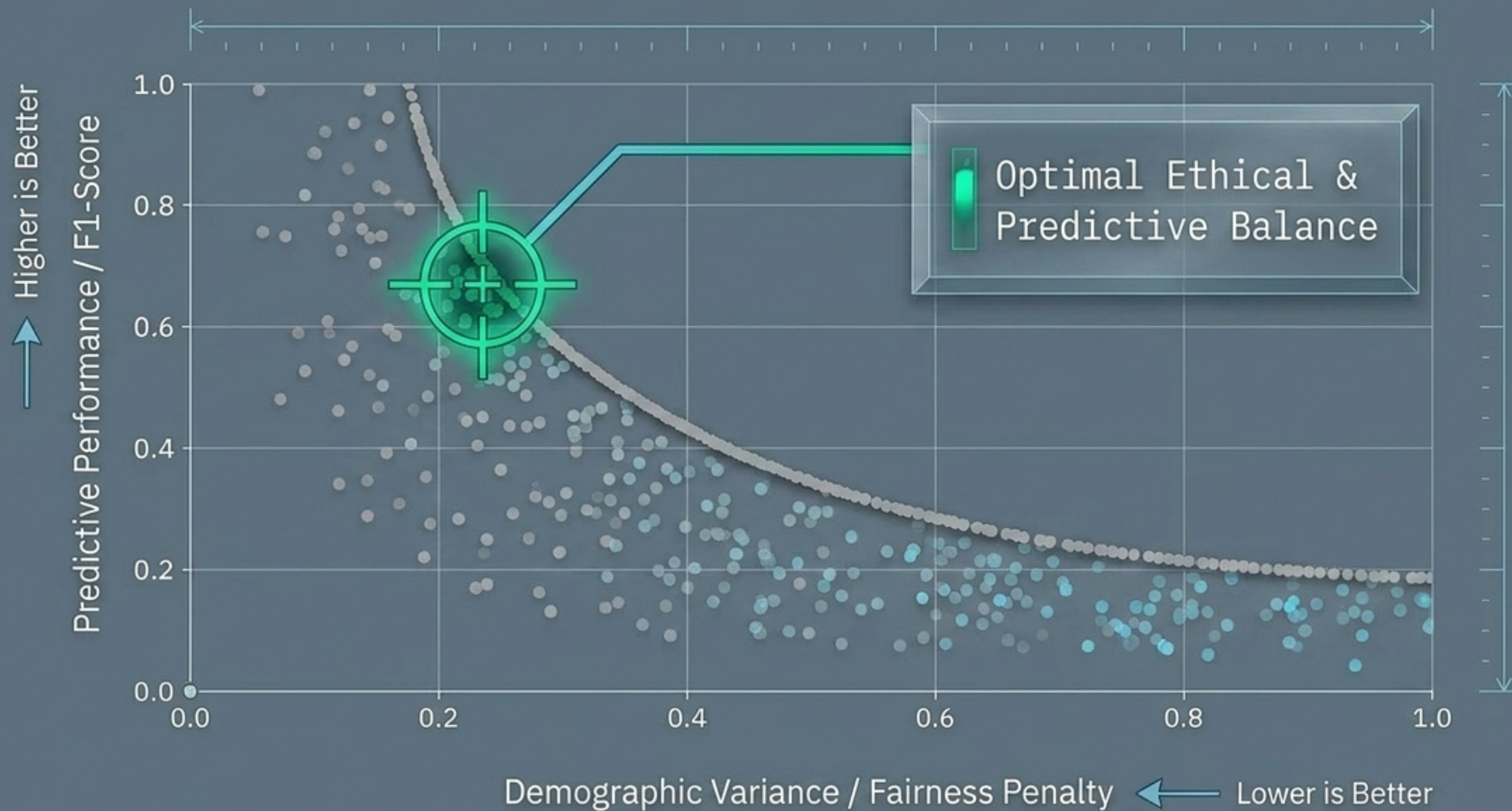
TabPFN Bayesian Transformer



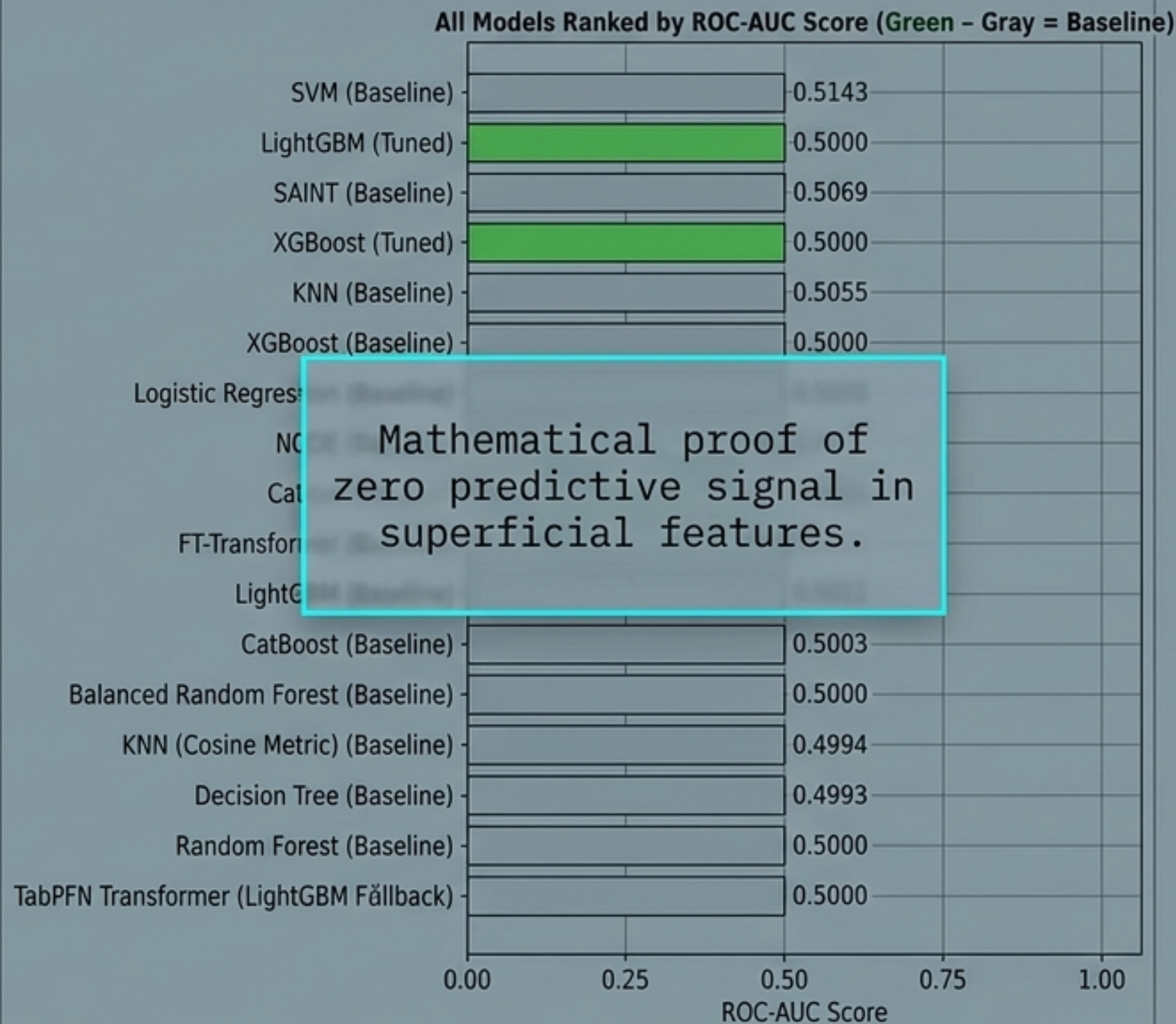
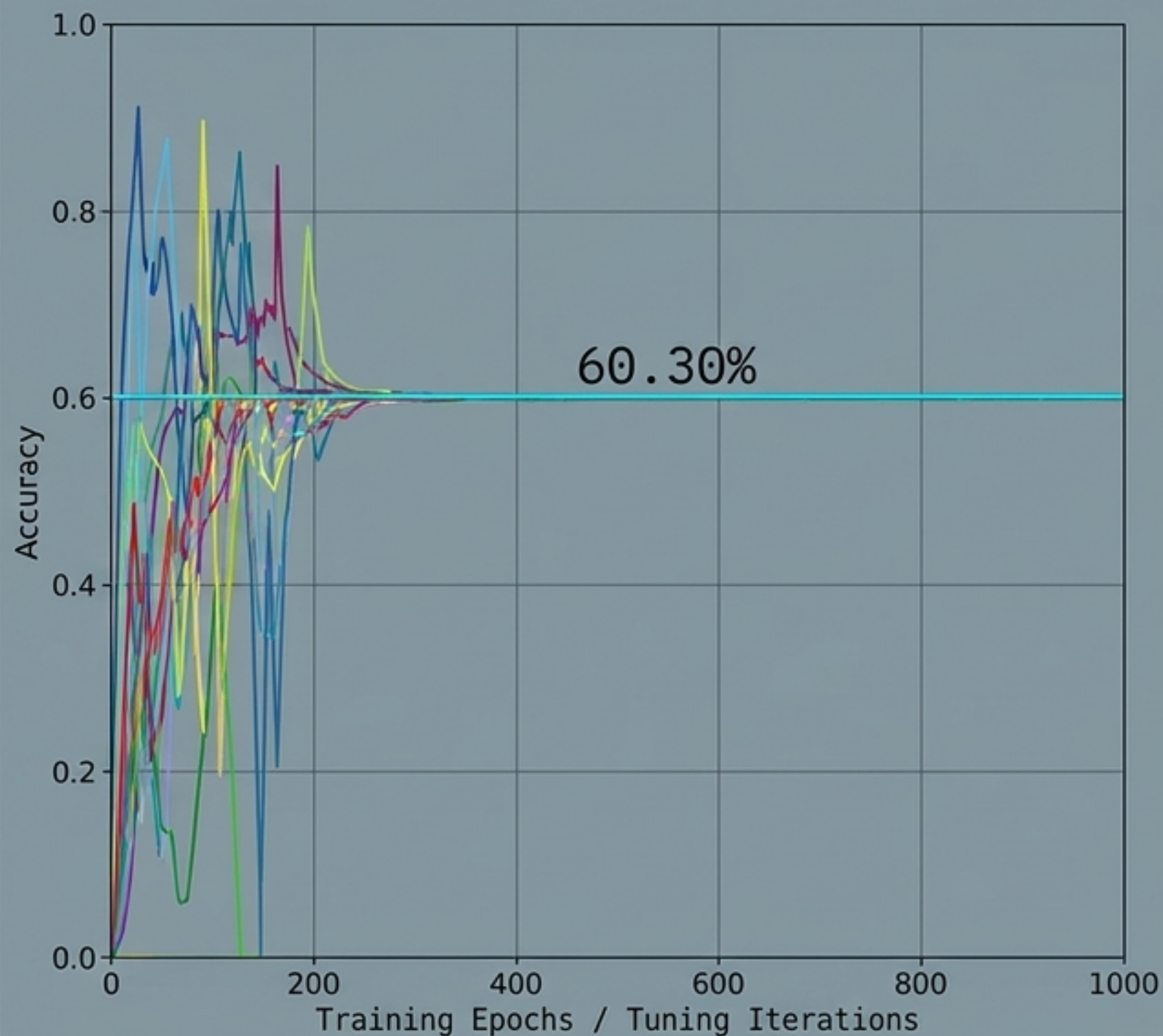
Brute-Force GPU Hyperparameter Optimization



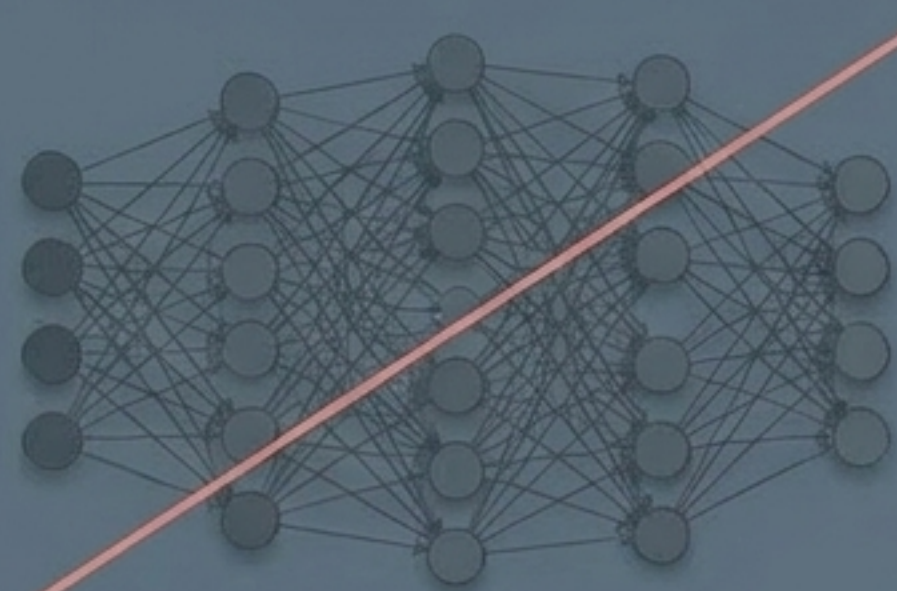
Multi-Objective Pareto Optimization



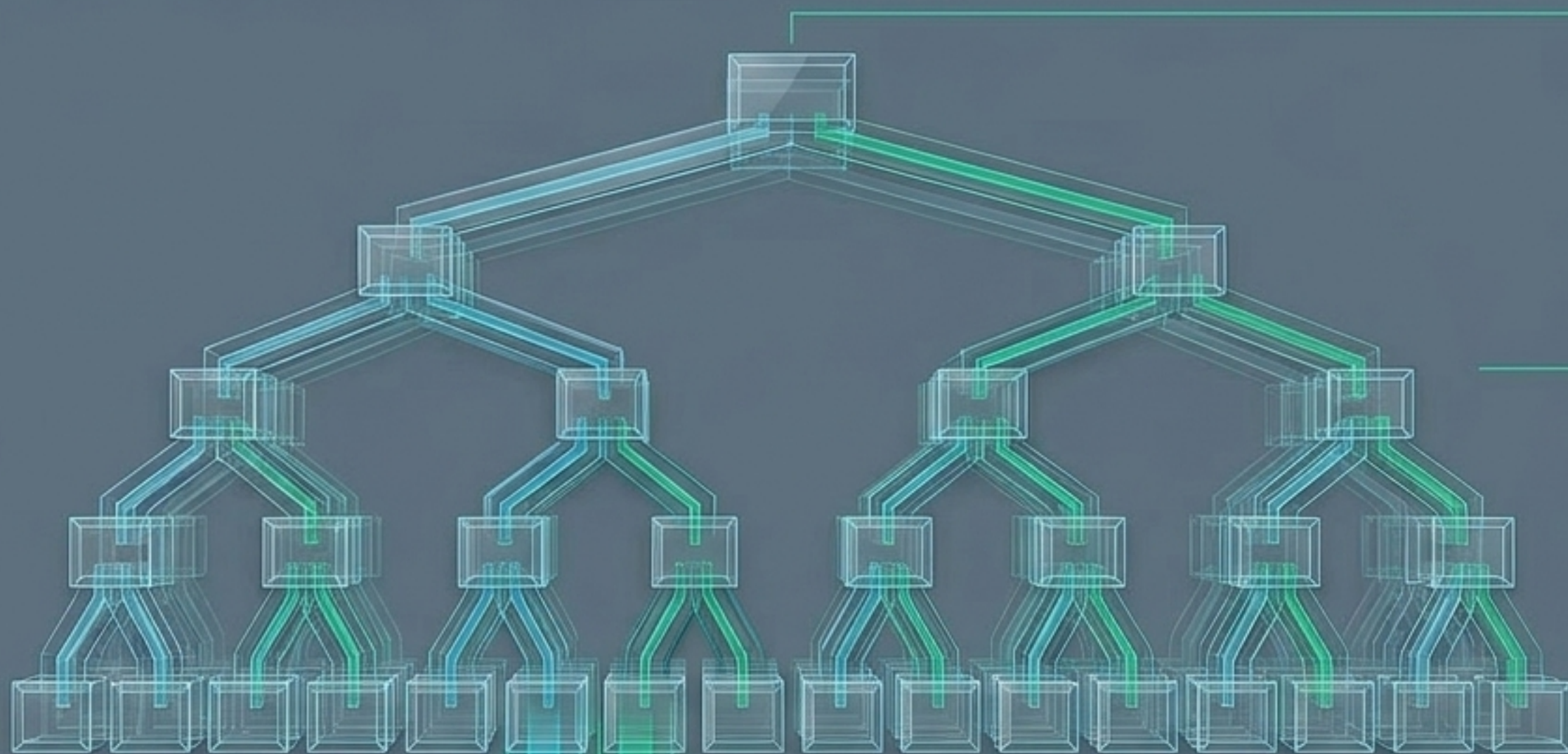
The Scientific Twist: The Absence of Predictive Signal



Champion Model Selection: The Glass-Box Mandate



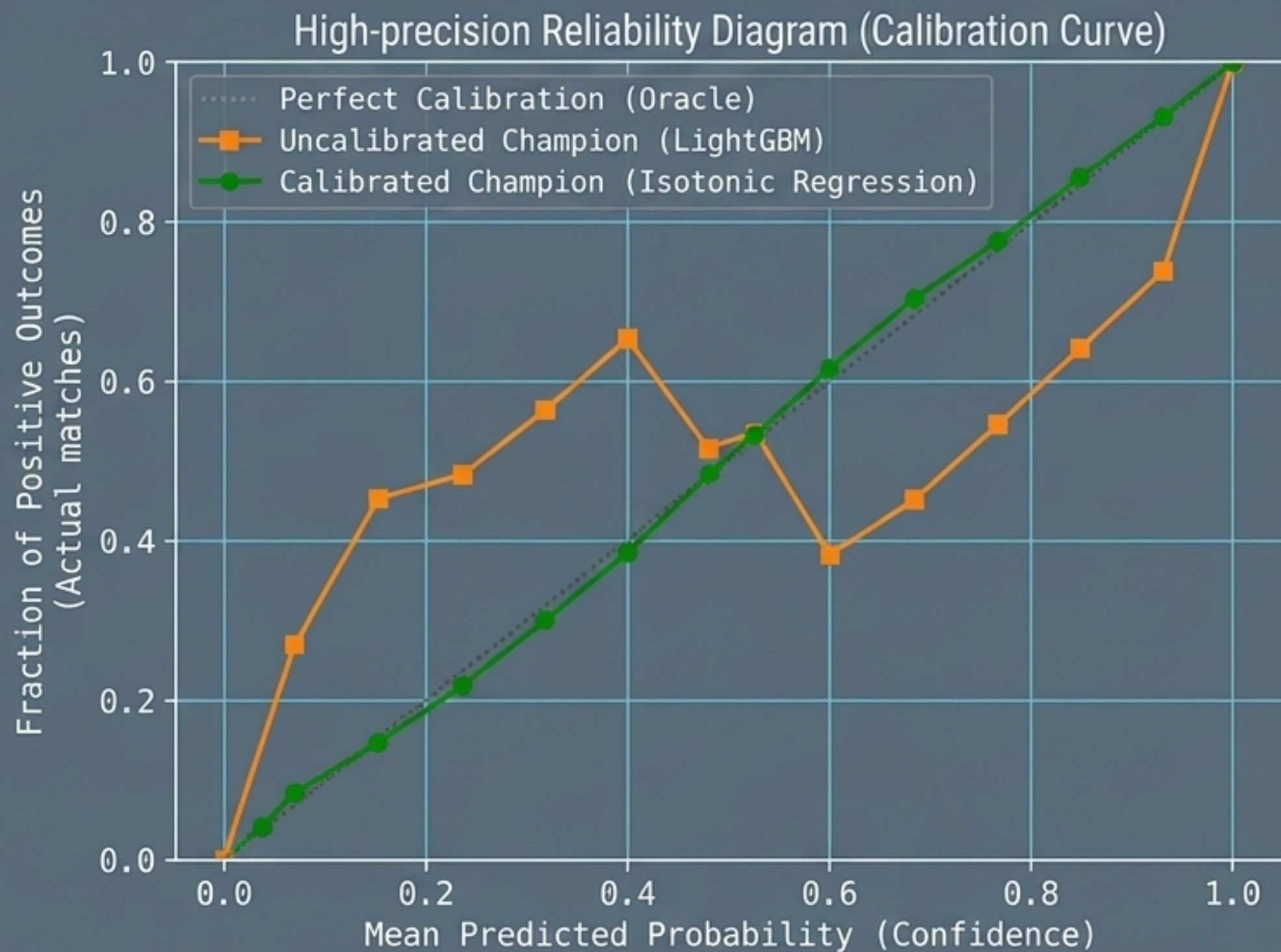
Opaque Neural Black-Boxes
(Computationally Wasteful)



Direct Explainability
(SHAP)

Actionable Recourse
(DiCE)

Probability Calibration for Trustworthy Deployment



✓
Brier Score
Minimized.
Honest Confidence
Levels Restored.